



Pennsylvania Department of Transportation State Safety Oversight Agency (PennDOT SSOA)

2024 Annual Report
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Acknowledgements

In accordance with Title 49 of the Code of Federal Regulations, Part 674.13(a)(7), attached is the Calendar Year 2024 Report of State Safety Oversight Activities for Rail Fixed Guideway Public Transportation Systems in Pennsylvania. The Pennsylvania Department of Transportation (PennDOT) State Safety Oversight Office developed the Annual Safety and Security Report as Pennsylvania's State Safety Oversight Agency (SSOA).

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SSOA State Safety Oversight Program

The Federal Transit Administration (FTA) delegates oversight of rail fixed guideway public transportation systems that operate with federal financial assistance authorized under 49 United States Code Chapter 53 to the states. This structure requires states to designate a state safety oversight agency (SSOA) to fulfill the requirements of the state safety oversight program, codified in 49 Code of Federal Regulation (CFR) Part 674, referred to as Part 674 hereafter.

In response to a National Transportation Safety Board (NTSB) investigation following a deadly Southeastern Pennsylvania Transportation Authority (SEPTA) accident in 1990, the Governor of the Commonwealth of Pennsylvania created the Rail Transit Safety Review Program (RTSRP) within the Pennsylvania Department of Transportation (PennDOT). The RTSRP is now known as the PennDOT SSOA. PennDOT SSOA has served as Pennsylvania's safety and security oversight for the state's rail transit agencies (RTAs) since 1991 – well before the FTA issued the first federal rule, 49 CFR part 659, in 1995, which went into effect in 1997.

On March 16, 2016, FTA issued the 49 CFR Part 674 final rule to replace 49 CFR Part 659. The 49 CFR Part 674 final rule standardized SSOAs across the country by requiring financial, legal, operational, and administrative independence from the transit agencies they oversee. Part 674 also enhanced investigative and enforcement authorities, training requirements, and required annual reporting to the Governor, the FTA, and the board of directors of the RTAs. The FTA certified Pennsylvania's SSOA under 49 CFR Part 674 on April 24, 2018. In 2024, the FTA issued one new final rule (Part 671) and three revised final rules (Parts 672, 673, and 674), making 2024 a major year for addressing safety throughout the industry. The new final rule, 49 CFR Part 671, establishes new safety standards for roadway worker protection (RWP) at all rail transit agencies. The revised 49 CFR Part 672 outlines new reporting requirements for federal safety training. The revised 49 CFR Part 673 requires heightened collaboration by agencies for conducting risk reduction assessments and coordinating safety committees. Finally, the revised 49 CFR Part 674 updated safety event reporting requirements. PennDOT SSOA integrated these changes into its 2024 Program Standard and revised supporting safety standards and standard operating procedures accordingly.

Information regarding the SSOA, along with a method to report an allegation of noncompliance to the SSOA, can be found on the PennDOT website at <https://www.penndot.pa.gov/Doing-Business/Transit/Pages/state-safety-oversight.aspx>. The SSOA issues its annual Program Standard with input from each of the RTAs it oversees. The 2024 Program Standard included major updates such as:

- Added requirements for RTAs to develop a Continuity of Operations Plan (COOP);
- Updated required qualifications and training for investigators;
- Updated minimum requirements for information that must be tracked in RTA hazard logs;
- Added information about PennDOT SSOA's risk-based inspection programs;

- Updated RTA and SSOA timelines for notifications and documentation related to triennial audits, inspections, field reviews, and CAPs;
- Added requirements for more frequent communication during the design phase of Safety and Security Certification;
- Added requirements for RTA safety and security committee meeting minutes to capture hazard log discussions and be shared with the SSOA;
- Updated organizational chart in Appendix H, which demonstrates the SSO Office as its own independent office;
- Implemented requirements from 49 CFR Parts 672, 673, and 674; and
- Implemented requirements from FTA Special Directive FTA 24-4-002.

The Pennsylvania SSOA oversees three RTAs: the Cambria County Transit Authority (CamTran) in Johnstown, Pittsburgh Regional Transit (PRT) in Pittsburgh, and SEPTA in Philadelphia.



Image 1. CamTran logo.



Image 2. SEPTA Metro logo.



Image 3. PRT logo.

The SSOA assumes oversight responsibility of these agencies and, as such, works with the agencies to continuously improve the implementation of their safety management systems (SMS) through activities such as technical assistance, safety and security procedural development, help in monitoring hazardous conditions, and investigations of safety events. The SSOA chairs a monthly cross-agency meeting to discuss SSOA activities and initiatives and to host the exchange of best practices and pertinent safety and security information throughout the Commonwealth. This is one way the SSOA fosters open communication between each RTA and itself, which will be key as PennDOT SSOA works towards implementing the federally mandated Risk-Based Inspection (RBI) Program across all the agencies, per Special Directive 22-44. During 2024, PennDOT SSOA continued to address FTA comments on its proposed RBI program language and details, and the SSOA ultimately received FTA approval of its RBI program in January 2025.

The PennDOT SSOA is also an active member and the 2024 Vice-President of the Transit Safety Oversight Association (TSOA), which is a non-profit organization built to serve professionals in the transit safety oversight field. TSOA's purpose is to support, encourage, develop, and enhance transit safety among SSOAs throughout the country through its national safety policy analysis, communication, education, and training opportunities. The PennDOT SSOA has overseen rail transit safety for over three decades. The SSOA continues to fulfill FTA-required program elements under Moving Ahead for Progress in the 21st Century, which encompasses 49 CFR Parts 672, 673, and 674 and provide advanced supplemental safety and security practices and standards in response to the unique aspects of Pennsylvania's RTAs.

Executive Summary

The 2024 Pennsylvania SSOA Program Standard applies to the CamTran Inclined Plane; PRT's Blue Line, Red Line, Silver Line, and Monongahela Inclined Plane; and SEPTA's Broad Street Line (B or Orange), Market-Frankford Line (L or Blue), Media-Sharon Hill Line (D or Pink), Norristown High Speed Line (M or Purple), Route 15 Trolley (G or Yellow), and Subway-Surface Lines (T or Green). As part of its jurisdiction, the SSOA also oversees PRT's MLK Jr. East Busway, South Busway, and West Busway, and SEPTA's Private Busway. Stations and maintenance facilities related to rail transit and busway operations are included within the SSOA's scope.

Throughout 2024, the SSOA worked collaboratively with CamTran, PRT, and SEPTA to continuously improve safety and security. The SSOA met with agency representatives through formal monthly, quarterly, and annual meetings. The SSOA also regularly attended meetings hosted by the RTAs to address and monitor safety and security concerns. SSOA oversight efforts included developing and issuing SSO Program Standards; reviewing and approving agency safety and security plans; fulfilling Part 672 training certification requirements; monitoring safety and security certification processes for capital projects; investigating events; refining the RBI program for federal compliance; conducting audits and inspections; working with agency personnel to oversee corrective action implementation; and engaging in discussions about security and emergency preparedness.

In 2024, the SSOA continued to update its triennial audit checklists to align with RTAs' Public Transportation Agency Safety Plans (PTASPs). These checklists will continue to undergo annual updates in conjunction with PTASP revisions. The SSOA also cemented its Data or Information for Hazard Identification (DIHI) process to identify potential safety hazards that, while compliant with the PTASP, could lead to injury, damage, or environmental harm. Throughout 2024, 102 DIHIs were identified at SEPTA. SEPTA tracked seven of the DIHIs as reportable hazards; six of the DIHIs created CAPs for ongoing tracking; and the remainder were either closed following SEPTA's completion of its initial hazard assessment form or are awaiting the completion of an initial hazard assessment form. At PRT, 30 DIHIs were identified in 2024 and 27 were closed after follow-up actions required under PRT's hazard management program.

Operations and stations safety inspections remained a key activity for the SSOA, with extensive reviews conducted at SEPTA and PRT, along each line at the agencies. Notably, the SSOA conducted 191 reviews of operational movements between SEPTA stations and 165 reviews of individual stations. The SSOA conducted 56 reviews of operational movements between PRT stations and 56 reviews of individual stations. Additionally, PennDOT SSOA continued the use of its light detection and ranging (LIDAR) device for speed reviews, with three total reviews conducted in 2024, comprised of one at SEPTA and two at PRT. In 2025, PennDOT SSOA plans to increase its number of LIDAR reviews and assess any increased needs for reviews.

The PennDOT SSOA also actively manages all RTA Corrective Action Plans (CAPs). Each RTA is required to submit CAP proposals to the SSOA for review and approval within 30 days of a

deficiency requiring corrective action being identified by the RTA. In 2024, SEPTA demonstrated significant improvements in managing open CAPs, having closed 137 CAPs throughout 2024, compared to 61 closed throughout 2023. SEPTA maintained 147 open CAPs at the end of 2024. Meanwhile, PRT maintained 15 open CAPs at the end of 2024, with 22 closed during the year; and CamTran had one open CAP at the end of 2024, with one closed during the year. CAPs and all subsequent updates and closure verifications are tracked through an electronic SMS database, facilitating streamlined management and automatic alerts. The SSOA is in the process of transitioning to a new electronic management system to more effectively process CAP approvals, extensions, and closures. The new system will provide better ongoing management of CAP statuses and contain tools that will be helpful in providing qualitative analysis of CAP information and ongoing monitoring of action items for each CAP. Additionally, the new system will streamline the CAP management process by utilizing automated memo creation and notifications for each stage of the CAP process.

Additionally, following major SEPTA safety events in the summer of 2023, the FTA initiated a Safety Management Inspection (SMI) in August 2023. Activities for Phase 1 of the SMI spanned September 2023 to December 2023, with FTA personnel traveling onsite for interviews, field reviews, and Roadway Worker Protection (RWP) training. The focus areas of the SMI were SEPTA's rail transit and bus operations and the effectiveness of rail transit oversight activities conducted by PennDOT SSOA. Both the PennDOT SSOA and SEPTA participated in FTA's onsite activities, follow-up virtual interviews, and the provision of requested documents. In 2024, Phase 2 commenced with the issuance of the FTA's SMI Report, which contained special directives that required the development and implementation of corrective actions to resolve FTA-identified deficiencies. PennDOT SSOA responded by developing CAPs, increasing its oversight activities, and coordinating heavily with SEPTA to address the FTA's concerns. In 2025, Phase 3



Image 4. Media-Sharon Hill Line car. September 20, 2024.

will consist of the FTA monitoring the implementation of corrective actions addressing the deficiencies identified through the SMI. The SSOA will allocate significant resources towards the closure of SEPTA SMI items through meetings with SEPTA and increased field presence. The SSOA will continue to provide monthly SMI updates to the FTA

regarding SEPTA's progress toward completing milestones for SMS implementation.

In 2025, the SSOA will also focus on enhancing its approved RBI program by conducting data-driven inspections per the RBI manual. The SSOA will continue to coordinate with RTA personnel to streamline data submittals and effectively utilize collected data to identify data-driven RBIs based on the prioritization process. The SSOA is also working with SEPTA on updating all the agency's Standard Operating Procedures (SOPs) and other documentation to ensure compliance with several major regulatory changes that FTA is issuing, including finalizing the RWP SOP update to include 49 CFR Part 671 requirements. In alignment with the regulatory updates requiring changes to RTA's RWP programs, the SSOA will coordinate with the covered RTAs to review and approve the RWP programs. These initiatives aim to bolster safety and security across all covered RTAs, ensuring continuous improvement and compliance with regulatory standards. Additionally, the SSOA will continue working alongside the covered RTAs for safety certification oversight, including oversight of SEPTA's Trolley Modernization Project and the Media-Sharon Hill Line communications-based train control testing project; PRT's automatic trip stop installation and stray current filtration; and CamTran's inclined plane rehabilitation and commissioning testing. Ongoing coordination with the RTAs to analyze hazard trends and assess risk through data modeling will be a focus of the SSOA's 2025 efforts. The SSOA will also share programmatic updates, procedural changes, and other PennDOT SSOA training opportunities with the RTAs during monthly meetings.

Federal Transit Administration (FTA) Coordination

Throughout 2024, the SSOA coordinated with the FTA through monthly calls to share programmatic updates on oversight activities; solicit guidance on regulations, standards, and



Image 5. PRT Silver Line trolley in operation to St. Anne's. July 18, 2024.

effective practices; review events; and discuss RBIs and relevant safety advisories and safety directives. These monthly calls were supplemented by the coordination activities between FTA, the SSOA, and SEPTA in connection with the ongoing SMI. The SSOA met regularly with the FTA as it developed its RBI program, incorporating the FTA's feedback at each milestone.

The SSOA also submitted the annual report to the FTA using the State Safety Oversight Reporting (SSOR) platform, which compiles safety data on SSO activities and records the SSO Program Manager's certification of compliance with applicable regulations. The partnership between the SSOA and covered RTAs, along with the FTA coordination,

continues to focus on achieving the highest levels of safety and security practicable, benefiting RTA patrons and employees and the public.

Oversight Program Activities

Establishment of PennDOT SSOA Office

In July of 2024, legislation was passed that formally created the State Safety Oversight Office for SSOA oversight activities within PennDOT. Prior to this legislation, the SSO Office was a division within PennDOT's Bureau of Rail, Freight, Ports, and Waterways. The Office is led by the Director, Elizabeth Bonini, and supported by two Regional SSOA Offices and two consulting firms. The Regional SSOA Offices are in the process of adding three staff positions this calendar year.

Ms. Bonini has managed the PennDOT SSOA since 2008 and is a World Safety Organization Certified Safety Executive, as well as a founding member and the 2024 President of the Transit Safety Oversight Association (TSOA). In addition, Ms. Bonini has a Transit Safety and Security Program (TSSP)-Rail Certification and holds training certificates that comply with the Public Transportation Safety Certification Training Program (PTSCTP), 49 CFR Part 672, as authorized by 49 U.S.C. § 5329(c)(1).

In May 2023, PennDOT filled two regional management positions to oversee safety and security in the eastern and western regions of the Commonwealth. Regional managers both hold their PTSCTP and have extensive rail transit experience and knowledge.

Safety Management Inspection (SMI)

From July 21 to August 2, 2023, SEPTA experienced numerous major safety incidents, including bus collisions, trolley collisions, and a trolley runaway/derailment event. On August 10, 2023, the FTA initiated an SMI through a notification to the PennDOT Secretary and SEPTA Accountable Executive. The PennDOT Deputy Secretary for Multimodal Transportation was designated by the Secretary as the SMI Point of Contact. Phase 1 of the SMI began in September 2023 and continued through the end of December 2023. The focus areas of the SMI were SEPTA's rail and bus operations and the effectiveness of oversight activities conducted by the PennDOT SSOA. The SMI activities of SEPTA's bus operations are outside the scope and authority of the PennDOT SSOA and are under the purview of FTA's authority.

The PennDOT SSOA and SEPTA participated in FTA's SMI activities (bus and rail), follow-up virtual interviews, and the provision of requested documents. FTA hosted an exit interview on January 30, 2024. In 2024, Phase 2 commenced with the issuance of FTA's SMI report with special directives that required the development and implementation of corrective actions to resolve FTA-identified deficiencies. PennDOT SSOA began implementing these corrective actions throughout 2024. In 2025, Phase 3 will focus on implementation monitoring, the activities of which will include announced and unannounced visits, document audits, CAP post-closure monitoring activities, and verification of SMS milestone completion.

Public Transportation Agency Safety Plan (PTASP)

Throughout 2024, the SSOA worked with the covered RTAs to ensure compliance with 49 CFR Part 673.11(a)(5), which requires that each transit agency conduct an annual review and update of the agency's PTASP. The 2024 SSOA Program Standard updates also provided the RTAs with further clarification of RBI program federal requirements and Safety Risk Management (SRM) process requirements. The SSOA dedicated significant work hours towards assisting RTAs with technical guidance in updating their PTASPs with compliant and practical language. In 2024, the SSOA approved both PRT's and CamTran's updated rail transit PTASPs, as well as SEPTA's rail transit PTASP, which underwent major updates in areas of SRM, employee safety reporting, safety committees, and SMS implementation throughout 2024. In 2025, PennDOT will continue to monitor the implementation of the PTASP in compliance with new requirements under 49 CFR Parts 671, 672, 673, and 674.

Safety Management Systems (SMS)

SMS implementation at each covered agency continues to be a focus of the PennDOT SSOA, including monthly and quarterly RTA meetings with standing agenda items to review status and progress. The SSOA supports RTAs in satisfying requirements promulgated by the FTA in accordance with Parts 673, including flexibility in the plan and progress at each RTA.

In 2024, RTAs provided positive updates on the status of SMS implementation through increased employee training; moving forward with the procurement of software designed to improve the collection and use of safety data for analysis and focused safety activities; and the promotion and enhancement of employee reporting systems. The PennDOT SSOA has continued to focus on SMS implementation during triennial audits and has continued to verify increased and more in-depth implementation of SMS activities during safety audits and inspections, especially following the release of the SMI. PennDOT analyzed major concerns identified during the SMI for applicability to the other RTAs under PennDOT's jurisdiction.

Hazard Identification

The SSOA regularly conducts unannounced inspections of operator activities and station conditions. These inspections are compiled into Operations and Stations Review Memorandums, which are sent to RTAs so they may correct any identified safety concerns and determine if there are systemic safety concerns. In 2024, the SSOA formalized use of its web-based reporting application to record operations and stations reviews, utilizing the resultant database of inspection information to identify trends and hotspots in operator performance and station conditions. This data has been used to compare to the internal reviews conducted by the RTA and triennial audits conducted by the SSOA for further identification of hazards, in addition to being integrated into the SSOA's risk prioritization process.

Document Updates

In 2024, the SSOA instituted a revised 2024 Program Standard to address areas such as revised CAP processes, the DIHI program, and ongoing evolutions of the RTAs' SMS processes. An additional key programmatic document that was finalized in 2024 was the PennDOT RBI Manual, which was submitted to the FTA and subsequently approved. This RBI Manual will be the guiding document for RBI development, planning, and rating moving forward. Additionally, in 2024 PennDOT updated each SOP in accordance with the respective update timeline and initiated the review of the RWP SOP to address 49 CFR Part 671 requirements (to be released in 2025).

Moreover, PennDOT reviewed and approved each RTA's revised PTASPs and Security and Emergency Preparedness Plans (SEPPs), which are required to be resubmitted following the Program Standard release. Additionally, PennDOT reviewed and approved Accident Investigation Manuals for both SEPTA and PRT. Finally, each year, the PennDOT SSOA reviews and approved the agencies' internal audit program and plan for the following year as well as their respective annual internal audit report.

SSOA Risk Based Inspection (RBI) Program

On October 21, 2022, the FTA issued Special Directive 22-44, which requires the SSOA to develop and implement an RBI program. The FTA issued a similar special directive to every SSOA across the country. In 2024, the SSOA developed its RBI Program Manual, an appendix to the Program Standard that documents RBI Program procedures. In addition, the SSOA conducted a workload assessment to demonstrate that the SSOA has sufficient staffing to conduct RBIs. The SSOA also revised its Technical Training Plan (TTP) to ensure that SSOA staff and contract support have sufficient knowledge to conduct RBIs. The SSOA coordinated with the RTAs under its oversight to ensure they could provide data across all safety-related activities for the SSOA's ongoing evaluation; the RTAs also acknowledged and added relevant provisions in the RBI Program Manual into their respective PTASPs. After review of all documentation outlined in this paragraph, the FTA approved the SSOA's RBI program on January 8, 2025.

PennDOT SSOA's RBI program uses data-driven decision-making, in addition to analysis of individual events, to assess the safety and security of RTA maintenance and operations practices. The SSOA will identify areas for RBIs in 2025, using data to proactively identify potential hazards that require further oversight or investigation. The program will ensure continuous monitoring of higher risk rules compliance with ongoing maintenance, operations, and safety activities.

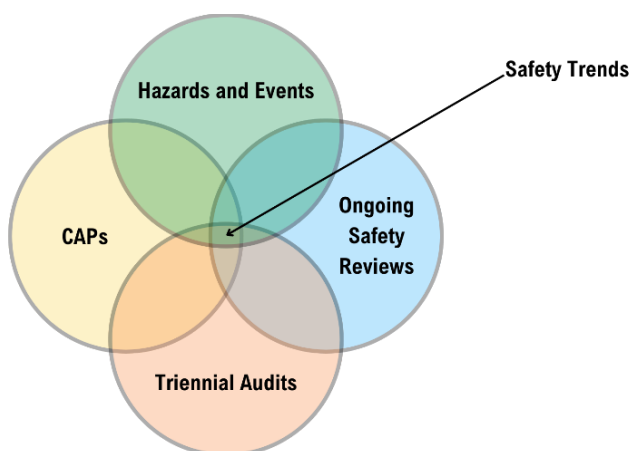


Figure 1. SSOA-Collected Source Data

As part of this data-driven decision-making, Special Directive 22-44 mandates that the SSOA collects data from the RTAs. In 2024, the SSOA coordinated with RTAs to determine available data sources. RTAs have committed to periodic submission of specified data deliverables concerning operations, maintenance, and safety. The type, frequency, and method of reporting for these deliverables has been codified in the RBI Program Manual.

In addition to RTA-supplied sources, the SSOA has continued to collect data and maintain its own internal databases based on SSOA-identified sources. The SSOA uses its internal databases to develop and track key performance indicators (KPIs), which are quantifiable measures the SSOA uses to evaluate the health and effectiveness of the SSO program. KPIs are tracked for each RTA, reported monthly and include metrics such as rules compliance violations, CAP statuses, triennial audit results, events analyses, near miss events, track fires, fatigue, and red signal violations. Figure 2 depicts excerpts from a sample 2024 monthly report. In 2025, the SSOA will continue to incorporate RTA-reported data into its monthly KPI report.

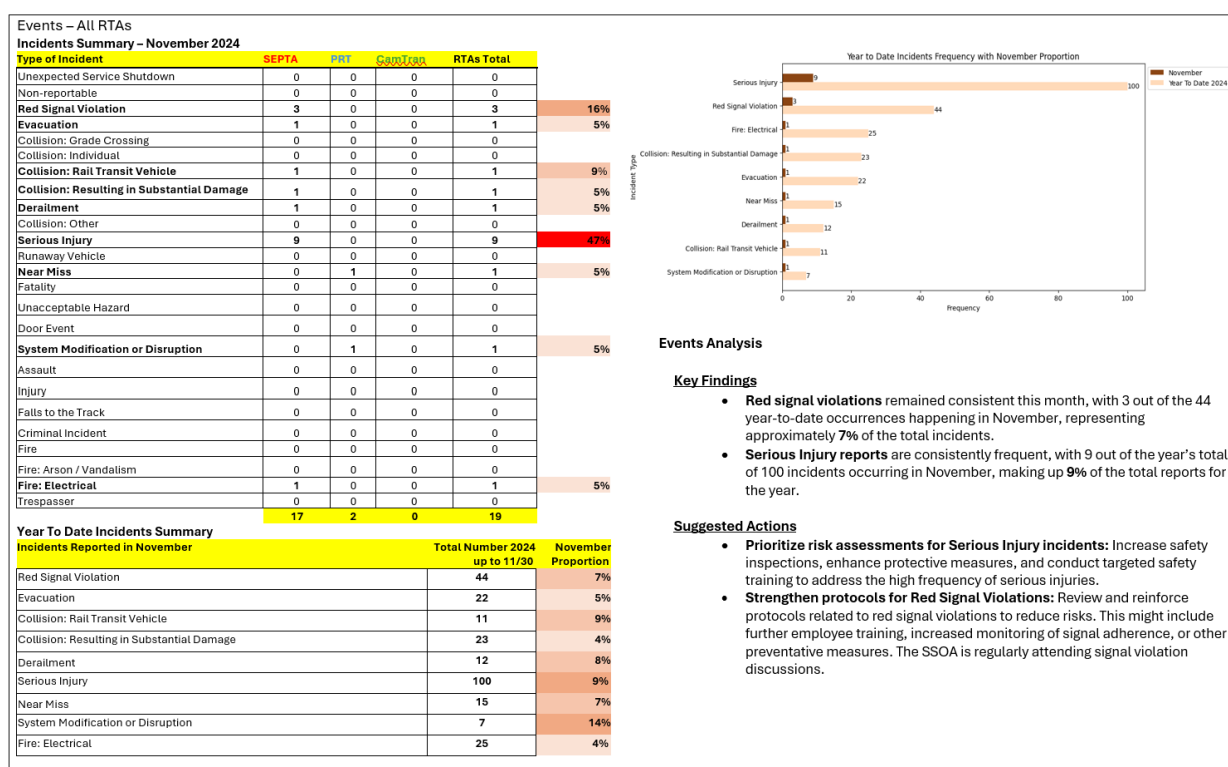


Figure 2. Excerpt from November 2024 Monthly KPI Report

Safety and Security Certification (SSC) and Capital Project Oversight

Each covered RTA is required to have a safety and security certification (SSC) process to ensure that hazards, threats, and vulnerabilities are adequately addressed prior to the initiation of passenger operations for new starts and major projects to extend, rehabilitate, or modify or replace existing systems, vehicles, or equipment. The SSOA is responsible for overseeing safety

certification at covered agencies and ensuring that SSC plans and documents submitted by an agency are aligned with the agency's program requirements. SSC oversight for projects falls under the criteria specified within the SSOA Program Standard and industry guidance. The SSOA recognized the importance of ensuring safety and security for the capital projects and created a formalized process that includes input from each RTA to complete a comprehensive assessment using a PennDOT SSOA checklist to evaluate capital projects for an initial SSC determination. In 2024, the SSOA oversaw the completion of SEPTA's Route 15 President's Conference Committee (PCC) Trolley restoration and return to revenue service. Current SSC projects being overseen by the PennDOT SSOA include CamTran's Inclined Plane rehabilitation; PRT's upgrade to its automatic trip stop system; SEPTA's implementation of communications-based train control on the Media-Sharon Hill Line (MSHL); and SEPTA's trolley modernization program.

PennDOT SSOA oversight activities include attending design and hazard analysis meetings and reviewing safety certification documentation such as safety and security certification plans; certifiable elements lists; reliability, availability, and maintainability plans; hazard logs; hazard analyses; and failure mode, effects, and criticality analyses. The SSOA also tracks progress by attending project meetings and SSC committee meetings and through status updates at monthly and quarterly meetings.

Safety Training

The SSOA has long incorporated safety training in its familiarization and training program for staff. Since the release of Part 672, the SSOA expanded on its training program and formalized many of the activities through the development of an SSOA Qualifications and Training Program (QTP), building upon the FTA-required TTP for each SSOA employee. The SSOA Program Standard requires training for both RTAs and the SSOA, including both federally mandated and PennDOT SSOA-specific training. The SSOA continues to offer and conduct its "SSO 101" and Safety and Security Certification familiarization training for RTAs.

Members of the SSOA must complete the PTSCPT and the TSSP training programs through the Transportation Safety Institute (TSI) within three years of their designation and must recertify every two years. Additionally, members of the SSOA team are required to complete specific training requirements to supplement the FTA-required training and familiarize new SSOA personnel with program-wide principles. In 2024, 11 SSOA members completed refresher training or received their initial certificates of PTSCPT completion. Furthermore, the SSOA continues to work with agency personnel to participate in RTA-specific training to become increasingly familiar with the unique aspects of each transit system. In 2024, the SSOA attended both initial and recertification courses for the RWP classes at PRT and SEPTA as well as supplementary RTA-led trainings, including the SEPTA Book of Rules Training. In 2025, the SSOA will continue efforts to attend and complete relevant safety, security, operations, and maintenance trainings at RTAs, especially as PennDOT SSOA begins oversight of the agency electric mass transit vehicle sticker program to ensure understanding of the RTAs' respective vehicle maintenance programs.

The SSOA also works with the RTAs to integrate the requirements for initial and refresher safety training for designated safety critical personnel into existing training programs at the RTAs. As an example of this effort, in 2024, the SSOA provided a tailored version of the SSO 101 familiarization training to SEPTA operations management personnel at the request of the agency's Chief Operating Officers. The SSOA will continue to provide programmatic training to internal staff and RTA staff as needed, including but not limited to go-bag training, operation and station review training, RTA physical characteristics training, National Transportation Safety Board guidance training, RWP oversight program, and Program Standard/SOP training.



Image 6. Market-Frankford Line trains at the 69th Street Yard. March 20, 2025.

Events

SSOA Investigation Process

In accordance with SSOA Program Standards and Part 674, the SSOA is responsible for the investigation of safety events, as well as the review of all hazard analysis and mitigation activities. With few exceptions, the SSOA delegates investigation authority to the covered RTAs, who conduct investigations for review and adoption by the SSOA. Similarly, the SSOA requires agencies to identify, report, and analyze hazards during transit operations.

The SSOA monitors a wide range of investigative activities, beginning with a notification of the event itself and culminating with the determination of a probable cause. Subsequent development of CAPs mitigate the possibility of recurrence. The SSOA and covered agencies work collaboratively throughout all stages of the investigation process. The SSOA receives an initial event notification from an RTA safety representative via an online form, at which point the SSOA begins to work alongside the agency investigator to conduct the investigation. The SSOA's process for adopting and closing an investigation includes reviewing the corresponding investigation report and ensuring that information is thorough and accurate. Investigations must include a probable cause and identify CAPs as appropriate. The SSOA provides ongoing reporting to all RTAs, which outlines new, adopted, rejected, and ongoing investigations.

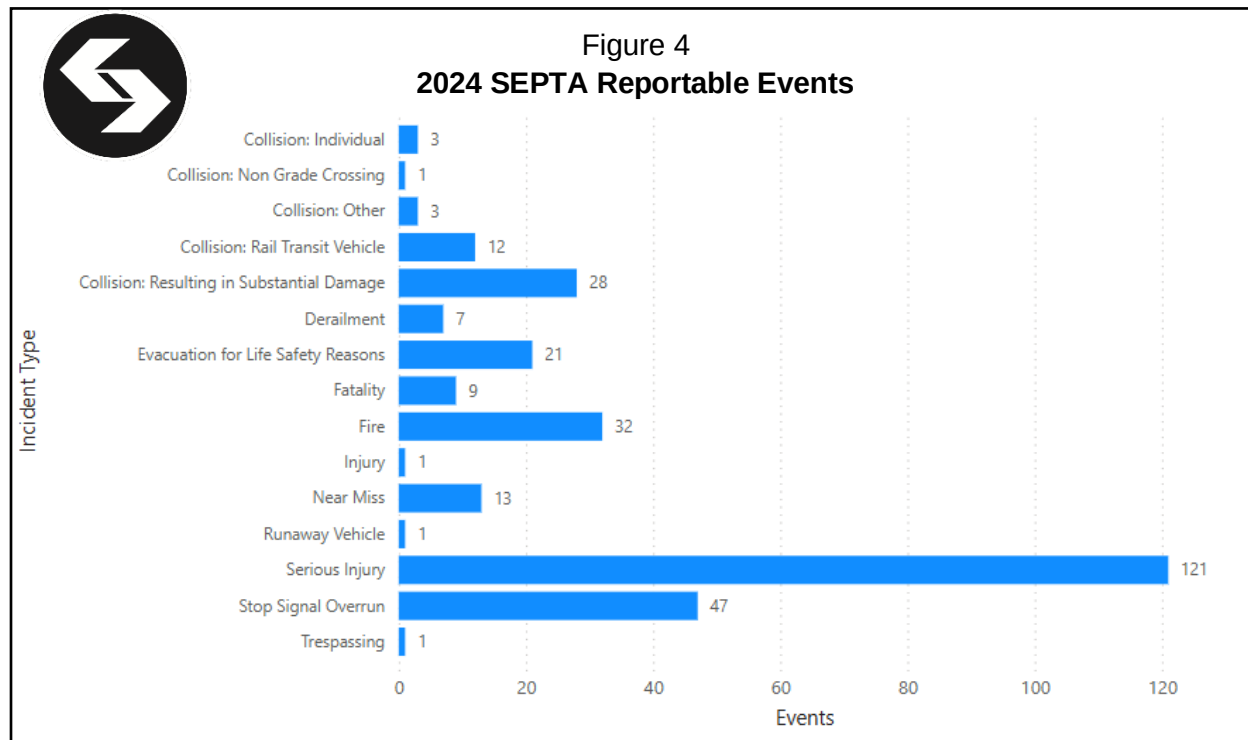
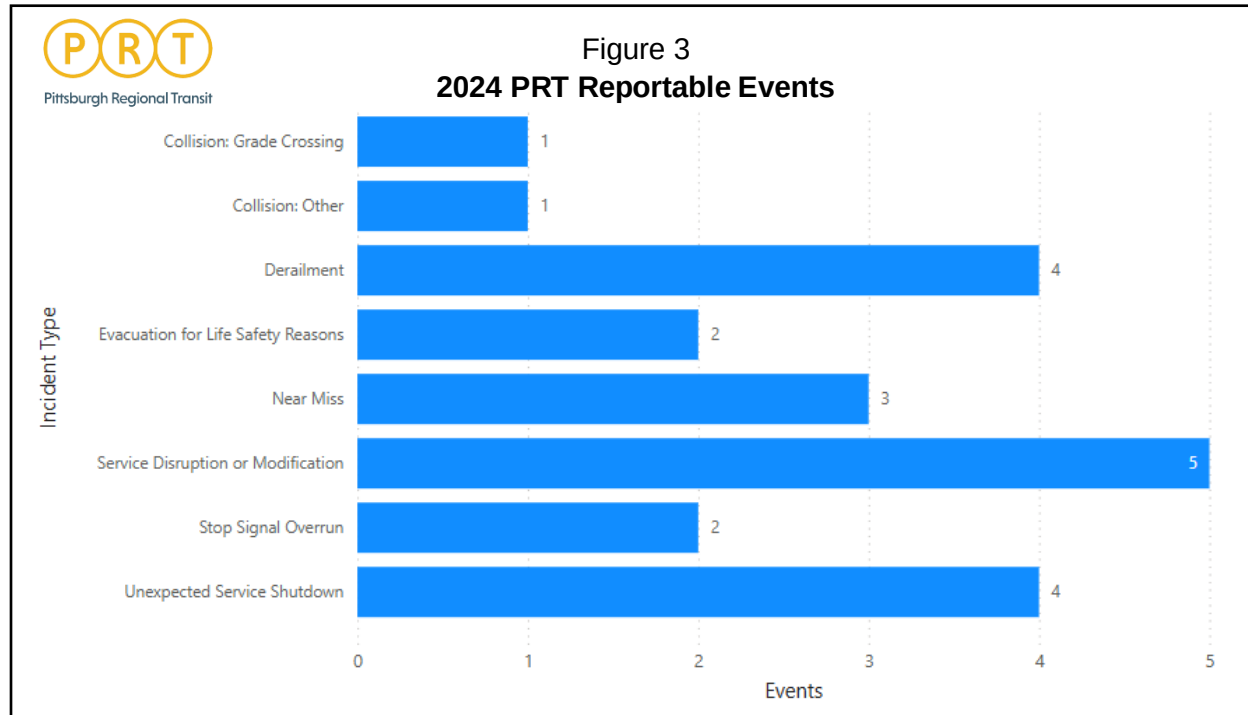
During 2024, the SSOA continued to utilize information gained from continual analysis of prior event data at SEPTA. The identification of trends for repeat events (such as the locations and time of day for incidents including close-clearance collisions and falls to the track) continues to aid SEPTA in the development of appropriate corrective actions and mitigations, and the SSOA in the determination of risk-based inspection activities. In 2024, the SSOA held three training workshops with SEPTA System Safety Division Accident Investigators to review safety event notification reporting requirements and SSOA safety event timelines and investigation report requirements, as prescribed in the Program Standard.

Note that the revision of 49 CFR Part 674 changed safety event reporting requirements and event reporting outcomes in 2025 reports. The SSOA updated its Program Standard to reflect the Part 674 updated requirements and coordinated with the RTAs to ensure they were aware of and properly aligned with the new requirements when reporting safety events to the SSOA and FTA.

2024 Reportable Events

In accordance with SSOA and FTA requirements, RTAs must report accidents to the SSOA within two hours and incidents within 24 hours of their occurrence. In 2024, PRT reported 87 percent of events within the prescribed reporting window, and SEPTA reported 93 percent of events within the prescribed reporting window. In 2024, the SSOA reviewed 313 event investigation reports for adoption from PRT, SEPTA, and CamTran.

Figures 3 and 4 on the following page depict the reportable event breakdown at PRT and SEPTA, respectively. Service Disruption or Modification was the most common reportable event at PRT, while Serious Injury was the most common reportable event at SEPTA.



2024 Occurrence Reporting

In 2024, the SSOA monitored occurrences at covered RTAs based on the reporting thresholds outlined in the SSOA Program Standard. Though many of the occurrences do not meet the threshold for FTA safety event reporting, occurrences are reportable to the SSOA in a monthly report. This rigorous reporting standard allows for trend analyses to support proactive mitigation and focused inspections of these occurrences before they result in more serious safety events.

Table 1 represents the agencies' reported occurrences to the SSOA in 2024. Most occurrences reported in 2024 were related to trespassing.

	CamTran	PRT	SEPTA	Total
Trespassing	0	137	867	1004
Collision: Other	0	1	0	1
Elevator/Escalator Injury	0	0	5	5
Unacceptable Hazard <i>As defined by the RTAs' PTASPs</i>	0	34	0	34
Door Event	0	0	0	0
Total	0	172	872	1044



Image 7. SEPTA Subway-Surface Trolley #9000 at the Elmwood Yard, Media. March 20, 2025.

Safety and Security Reviews

Triennial Audits

The SSOA performs triennial safety and security audits at each of its covered RTAs on an ongoing basis over a three-year cycle. In 2024, the SSOA completed its 2022-2024 triennial audit cycle. Triennial audits include a comprehensive document and record review, interviews, and field observations when warranted. SSOA audits are informed by Pennsylvania requirements, federal regulations, and effective industry practices. Audits focus on compliance with each RTA's procedures and plans. The triennial audit process is outlined in Figure 5.

The process by which audits are conducted remains consistent with prior years; however, specific topics are audited under the SMS framework. The SSOA continued SMS-based auditing in 2024 and released updated triennial audit checklists for each audit using the PTASP review checklist as a foundation. The utilization of SMS-focused checklists allows the SSOA to perform Part 674-compliant audits and assist the covered RTAs in identifying any gaps related to SMS; it also ensures each RTA is performing in accordance with their PTASP. Ahead of audits, the SSOA team reviews various materials related to the topic: past audits, inspections, opened and closed CAPs, DIHIs, and observations related to the audited topic. The SSOA then conducts safety assurance evaluations during the audit to ensure that corrective actions implemented to mitigate previously noted concerns have remained in place and there are no repeat safety concerns.

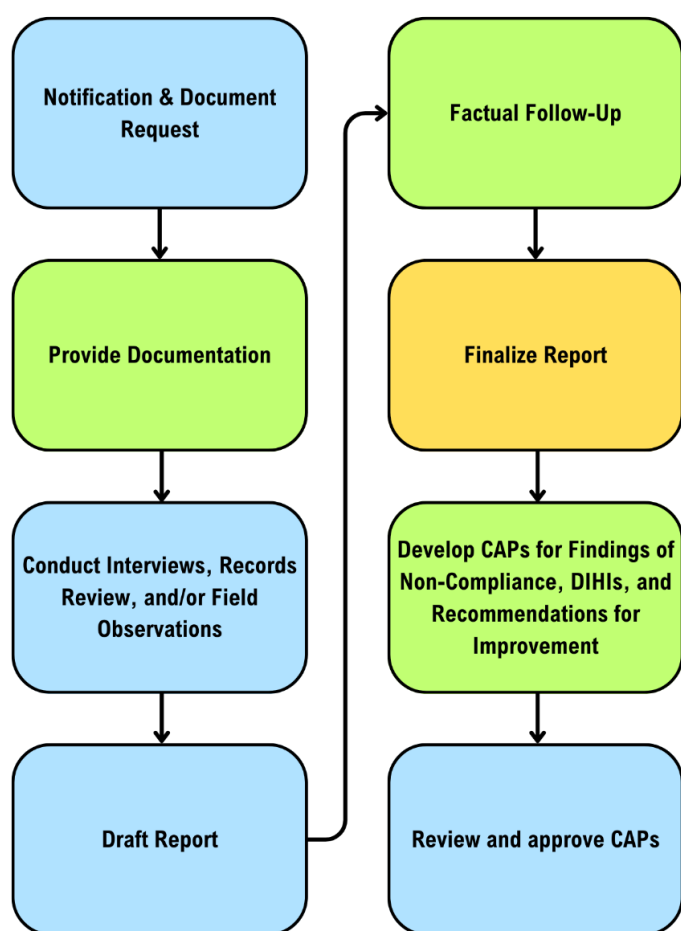


Figure 5. Triennial Audit Process

Audits by the SSOA may cover PTASP and SEPP implementation broadly, or they may focus on a narrow range of issues of a more technical matter based on previous audits. In 2024, the SSOA performed eight safety triennial audits at PRT, covering the following topics: Drug and Alcohol and Fitness for Duty; Communications; Light Rail Vehicle (LRV) Maintenance; PTASP Safety Management Policy; Non-Revenue Vehicle Maintenance; Stations and Busways maintenance

and inspection; Management of Change; and Occupational Safety and Hazardous Materials. The SSOA also conducted one security triennial audit at PRT in 2024 covering SEPP compliance.

The SSOA completed seven safety triennial audits at SEPTA, covering the following topics: Vertical Transportation; Double-Ended Light Rail Vehicle (DELRV) and Norristown High Speed Line (N5) Vehicle Maintenance and Inspection; Drug and Alcohol Policy and Testing; President's Conference Committee (PCC) Vehicle Maintenance and Inspection; Single-Ended Light Rail Vehicle (SELRV) Maintenance and Inspection; Safety Management Program (SMP) Policy Statement and Safety Assurance Program; and System Modification and Safety Certification Program. The SSOA conducted two security triennial audits at SEPTA in 2024, covering Safety and Security Effectiveness Preparedness Plan (SSEPP) Elements 1-4 and SSEPP Element 5. The SSOA also completed three special safety audits as part of the FTA SMI; these audits were of SEPTA's Operations Control Center (OCC), SEPTA's Broad Street Line emergency callboxes, and SEPTA's on-track safety program.

The SSOA put triennial audits of CamTran on hold until the system reopens; however, CamTran requested the SSOA perform a Safety Certification and Change Management audit in 2024, which the SSOA conducted in June 2024. A 2025-2027 audit cycle is scheduled and oversight of the SSC is ongoing.

In addition to performing triennial audits, the also SSOA attends and monitors internal safety and security audits at each covered agency and tracks CAPs that arise from these internal audits conducted by the RTA. In 2024, the SSOA oversaw or reviewed 18 internal audits at PRT and 23 at SEPTA. Like triennial audits by the SSOA, internal audits of CamTran were put on hold until the system reopens, so CamTran did not conduct audits in 2024, but a 2025-2027 internal audit cycle was developed.

Table 2 shows the 2024 audit activity at each RTA and Table 3 shows the total number of Findings of Non-Compliance from SSOA audits.

Table 2 Audits Performed in 2024			
	CamTran	PRT	SEPTA
SSOA Safety Audits	1	8*	10**
SSOA Security Audits	0	1	2

*Two of the eight safety audits were performed in 2024, and the final reports were issued in 2025.

** This total includes FTA SMI special safety audits. Three of the 10 SEPTA safety audits were performed in 2024, and the final reports were issued in 2025.

Table 3 Findings of Non-Compliance from Audits Performed in 2024			
	CamTran	PRT	SEPTA
SSOA Safety Audits Findings	2	12	47*
SSOA Security Audits Findings	0	0	1

*This total includes Findings of Non-Compliance from triennial safety audits and audits conducted through the FTA SMI.

Triennial Audit Checklist Updates

In 2024, the SSOA continued to update its triennial audit checklists to better align with the RTAs' PTASPs. The SSOA continued to review each section of the agencies' updated PTASPs to ensure that every aspect is included in agency-specific checklists. Additionally, the SSOA incorporated further information regarding SMS implementation into the checklists. The checklists will be updated annually when agencies update their PTASPs.

Data or Information for Hazard Identification (DIHI)

For each covered RTA, the SSOA tracks DIHIs, which refers to data and/or information that is technically compliant with an RTA's PTASP but has been identified to present a real or potential condition that can cause injury, illness, or death; damage to or loss of the facilities, equipment, rolling stock, or infrastructure of a public transportation system; or damage to the environment. Once a DIHI is reported, the RTA must use their documented safety risk management process to determine if a hazard has been identified and conduct a subsequent safety risk assessment. If the safety risk assessment determines a hazard rating that is reportable to the SSOA per the Program Standard, then the RTA will be required to follow the CAP process. All other identified hazards will be mitigated and monitored through the RTA's safety risk management process outlined in the PTASP.

Table 4 shows the total DIHIs that were identified in 2024 at each RTA as of December 31, 2024, as well as the total DIHIs that were closed in 2024. PennDOT and SEPTA are working diligently to close the open portion of DIHIs.

Table 4 DIHIs as of December 31, 2024				
	CamTran	PRT	SEPTA	Total
Identified in 2024	0	30	102	132
Closed in 2024	0	27	55	82

SSOA Operation and Station Inspections

The SSOA operations and stations safety inspections are an essential part of the SSOA ongoing safety oversight activities to observe compliance with applicable safety rules and identify potential hazards. The SSOA conducted operations and stations reviews at covered modes at SEPTA and PRT throughout 2024. The data and observations collected by the SSOA during operations and stations reviews are consolidated in a database and the SSOA analyzes the operations and stations review data for trends; this effort will help to inform the SSOA's risk-based inspections. The SSOA sent all observations made during operations and stations reviews to the RTAs via electronic memos. Any critical safety concerns are communicated immediately with agency safety personnel, control center personnel, or agency police, according to the Operations and Stations Review SOP. Additionally, in 2024, the SSOA continued use of a LIDAR speed detection device and trained additional team members to use it for speed reviews on covered modes at SEPTA and PRT.

Of note, as part of SEPTA's SMS implementation, SEPTA created an email list-serve with staff responsible for the review of these SSOA reports so any identified hazards are addressed immediately as appropriate.

Table 5 shows the total operations and stations reviews completed for PRT and SEPTA in 2024.

Table 5 Operations, Stations, and LIDAR Reviews Performed in 2024		
	PRT	SEPTA
Operation Reviews	56	191
Station Reviews	56	165
LIDAR Reviews	2	1

Corrective Action Plans (CAPs)

Covered RTAs are required to develop CAPs for various deficiencies and hazards identified through triennial audits, event investigations, hazard reports, rules compliance monitoring, and inspections. All CAPs must be submitted by the RTA to the SSOA for review and approval within 30 calendar days of an identified deficiency. Subsequently, the SSOA is responsible for reviewing, approving, and verifying implementation. The CAP process is depicted in Figure 6.

The SSOA reviews the status of CAPs with RTA safety and security personnel and other responsible departments, as appropriate, on an ongoing basis, but also works with RTAs at least monthly to update all CAPs. During these updates, RTA and SSOA personnel review progress towards the closure of open CAPs and identify documentation required to verify implementation. Verification may consist of field reviews, record reviews, photographs, and/or plan, policy, or procedure revision. Tables 6, 7, and 8 display the number of open and closed CAPs per year for CamTran, PRT, and SEPTA, respectively.

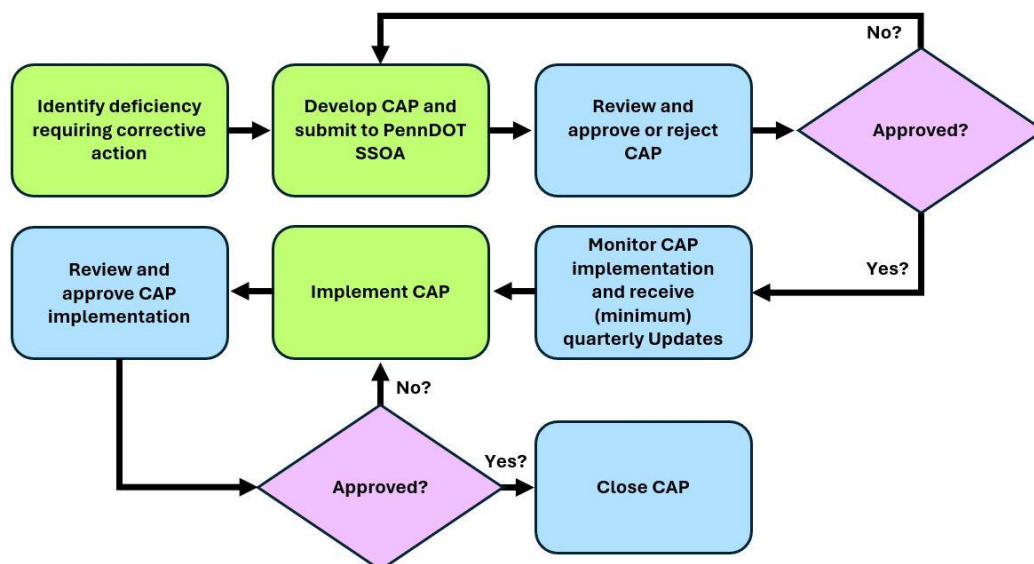


Figure 6. Corrective Action Plan (CAP) Process

 Table 6 CamTran CAPs by Year as of December 31, 2024							
	2019 & Before	2020	2021	2022	2023	2024	Total
Open	0	0	0	0	0	1	1
Closed in 2024	0	0	0	0	0	1	1

As of December 31, 2024, CamTran had one open CAP and closed one CAP in 2024.

 Table 7 PRT CAPs by Year as of December 31, 2024							
	2019 & Before	2020	2021	2022	2023	2024	Total
Open	1	0	1	0	3	10	15
Closed in 2024	0	0	0	2	14	6	22

As of December 31, 2024, PRT had 15 open CAPs and closed 22 CAPs in 2024.

 Table 8 SEPTA CAPs by Year as of December 31, 2024							
	2019 & Before	2020	2021	2022	2023	2024	Total
Open	7	0	8	7	57	68	147
Closed in 2024	5	0	12	21	68	32	137

As of December 31, 2024, SEPTA had 147 open CAPs and closed 137 CAPs in 2024.

The SSOA tracks all CAPs for each covered transit agency through an electronic SMS database. This database allows the SSOA to track CAP information and updates in a streamlined manner, as well as manage all forms of electronic verification materials, perform analyses, and disseminate automatic alerts surrounding upcoming closure. From the SMS database, the SSOA and the RTAs can also review KPIs of CAPs to assess areas where attention may need to be

focused. The hazard ratings of open CAPs as of December 31, 2024, are depicted in Table 9, and the sources of open CAPs are shown in Table 10.

Table 9 Hazard Ratings of Open CAPs as of December 31, 2024				
	CamTran	PRT	SEPTA	Total
High	0	0	1	1
Serious	0	0	13	13
Medium	0	3	119	122
Low	1	12	14	27
Total	1	15	147	163

Table 10 Source of Open CAPs as of December 31, 2024				
	CamTran	PRT	SEPTA	Total
RTA Internal Audit	0	1	13	14
SSOA Audit	1	12	17	30
Reportable Hazard	0	1	13	14
Accident/Incident	0	1	76	77
SSOA Inspection	0	0	23	23
FTA SMI	N/A	N/A	3	3
Other*	0	0	2	2
Total	1	15	147	163

*Any other hazard or safety concern that requires corrective action.

Security and Emergency Preparedness Efforts

Although the FTA relinquished its security oversight with the release of Part 674, they retained emergency preparedness as an SSO federal responsibility as part of each agency's PTASP. The SSOA has opted to maintain security oversight of its covered agencies, while working with the TSA as necessary to ensure that SSOA requirements do not conflict with those of the TSA. The SSOA continues to perform triennial audits of the RTA's SEPP and participates in any internal security audits, threat and vulnerability assessments (TVAs), drills and exercises, and other security and emergency preparedness related items. In 2024, the FTA released 49 CFR Part 673 for increased focus on mitigating transit worker assaults, which is loosely related to transit security processes. PennDOT will assist in oversight of transit worker assault mitigation via audits of the PTASPs.

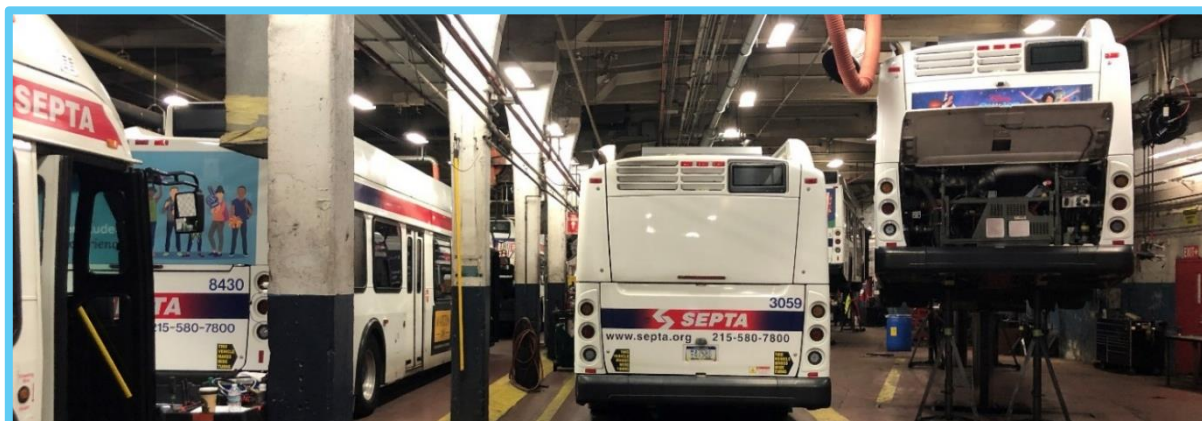


Image 8. Callowhill Bus Maintenance Facility. April 25, 2024

System Security and Emergency Preparedness Plan (SSEPP)/Security and Emergency Preparedness Plan (SEPP)

The SSOA received the 2024 revisions of the System Security and Emergency Preparedness Plan (SSEPP) for SEPTA and the SEPPs for PRT and CamTran. In accordance with the SSOA Program Standards, the SSOA reviews the plans and either approves or rejects them for use. The SSOA compared the RTAs' SSEPP/SEPPs with the SSOA Program Standard and applicable checklist created from tools provided by the FTA. After SSOA internal review, the SSOA found that each plan met current requirements and approved them for use. The SSOA approved each RTA's SSEPP or SEPP on the following dates:

- CamTran: PennDOT SSOA approved the SEPP on October 24, 2024
- PRT: PennDOT SSOA approved the SEPP on October 23, 2024
- SEPTA: PennDOT SSOA approved the SSEPP on October 31, 2024

Looking Forward

In 2025, the SSOA will continue to work closely with its covered RTAs regarding safety and security initiatives. The SSOA's focus areas for 2025 and beyond include the following:

RBI Program Development. The SSOA submitted its RBI Program Manual and supporting documentation to the FTA in 2024, which was subsequently approved in January 2025. In compliance with Special Directive 22-44, the SSOA will demonstrate six months of compliance with its RBI Program by January 2026. In this six-month compliance period, the SSOA will conduct the first announced and unannounced RBIs of RTAs under its oversight. PennDOT will continue to coordinate with RTA personnel to receive streamlined data submittals to best prioritize data-driven RBIs. Based on knowledge gained during this implementation period, PennDOT will also continue to develop procedures and best practices for RBIs.

RWP Oversight. In 2025, PennDOT intends to finalize its revised RWP SOP to address new internal requirements and requirements from the new regulation: 49 CFR Part 671. Additionally, PennDOT will plan to start conducting the annual auditing of the RTAs' RWP compliance programs as well as monitor each RTA's internal compliance check programs. During 2025, PennDOT will work with each agency to review and approve their respective RWP Programs and Manuals. Additionally, in 2024, PennDOT conducted an audit of the RWP program at SEPTA as a part of the SMI. PennDOT will conduct ongoing monitoring from the findings resulting from this audit report via the CAP process.

Safety Assurance. The PennDOT SSOA has been actively working with the RTAs on the regular self-identification of deviations, non-conformities, and performance to oversee safety and effectively impact their safety culture.

SMI Closure. PennDOT SSOA has submitted closeout proposals to the FTA for approval. Through collaboration with SEPTA, the development of CAPs, and implementation monitoring oversight plans, the SSOA has entered a steady routine of activities that strive toward continuous improvement. PennDOT and SEPTA will continue to work with the FTA to address SMI findings.

SMS Implementation. The SSOA continues to update the FTA monthly on SEPTA's SMS Implementation progress as well as oversight activities performed to verify each completed milestone. SEPTA's progression in launching a new safety-focused data analysis, risk mitigation, and employee reporting tool will enhance the effectiveness of SMS.

Streamline Reporting Mechanisms. The SSOA will continue to monitor and review incoming events and hazard data, with improvements to timeliness and quality, through the provision of electronic forms accessible by each RTA. An increasing amount of safety

data is being transferred to a centralized cloud-based database, enabling real-time access and analysis for event reporting, monthly monitoring, CAP reviews, and RBI prioritization. The SSOA is working to further expand its data gathering abilities through 2025 and beyond.

Safety Certification Oversight. The SSOA will continue its ongoing monitoring of safety certification for relevant projects across the RTAs. At SEPTA, key projects under SSOA certification review include: the trolley modernization design and planning and the MSHL CBTC testing, trial operations, and entry into revenue service. PennDOT is now monitoring each upcoming project at SEPTA to further identify projects that may need PennDOT review for concurrence regarding SSC. At PRT, the SSOA is monitoring the ongoing installation of the automatic trip stop system including a current filtration system for stray current identified in 2024. At CamTran, the SSOA will provide ongoing support to review the first article testing of manufactured components for the inclined plane and supporting and reviewing the commissioning testing planned for 2025.

RTA Coordination. The SSOA will continue its field presence and coordination with RTAs and look to expand SMS implementation within the agencies. For the SSOA, this will include analyzing hazard trends, assessing risk through data modeling, examining the connection between events and CAPs, and determining best practices regarding CAP closure to share with the other RTAs. Further, the SSOA will engage in more hands-on knowledge sharing with the RTAs, such as an exchange where the SSOA provides technical assistance to the RTAs and in turn the SSOAs attend more technical training at the RTAs. Moreover, the SSOA will continue to work with the RTAs to assess committee structures and ensure compliance with procedures and requirements from 49 CFR Part 673. Finally, the SSOA aims to finalize the RWP SOP and plan discussions with the RTAs in 2025. In 2025, PennDOT will further monitor and review the completion of Form J's at each agency to assess the necessity of SSC completion. Additionally, during monthly RTA calls, PennDOT will share programmatic updates, procedural changes, and other PennDOT SSOA training opportunities. In 2025, the PennDOT Electric Mass Transit Vehicle Inspection Program will fall under the SSOA office. This will involve agency maintenance training and further inspection data monitoring by the SSOA throughout 2025 and beyond.

Update SSOA Documentation. The SSOA will continue to update its Organization Chart, Program Standard, SOPs, including the RWP Program. As needed, PennDOT may need to further revise and clarify its procedures and will do so as required.



*Image 9. CamTran Inclined Plane. May 1, 2025.
This and all other photos in report taken by PennDOT SSOA.*

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